

HW 06: Plots and Explanations

ECE 0402: Signals, Systems & Probability

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1) Problem 1

The signal is

$$x(t) = e^{-t}, \quad 0 \leq t < T, \quad T = 3$$

with periodic extension of period $T = 3$. The fundamental angular frequency is

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{3}$$

1.1) Plot of the signal

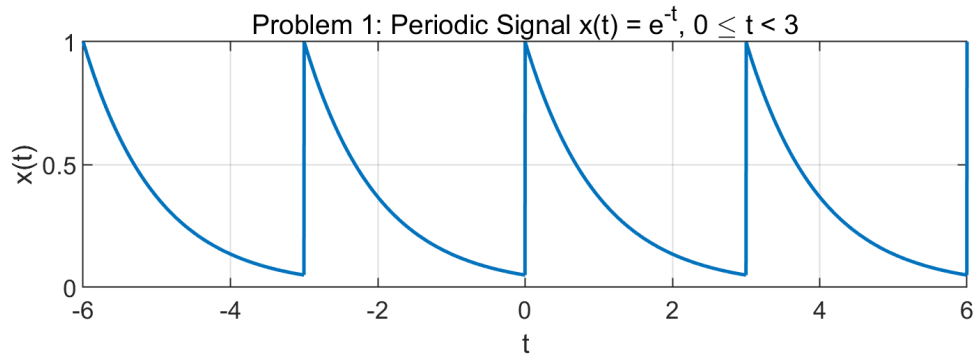


Figure 1: Problem 1 signal plotted over multiple periods.

1.2) Theoretical Fourier series coefficients

Using the complex exponential Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

the coefficients are

$$a_k = \frac{1}{T} \int_0^T e^{-t} e^{-jk\omega_0 t} dt$$

Substituting $T = 3$ and $\omega_0 = \frac{2\pi}{3}$ gives

$$a_k = \frac{1}{3} \int_0^3 e^{-(1+jk\omega_0)t} dt = \frac{1 - e^{-3}}{3(1 + jk\omega_0)}$$

Therefore,

$$a_k = \frac{1 - e^{-3}}{3 + j2\pi k}$$

In particular,

$$a_0 = \frac{1 - e^{-3}}{3}$$

1.3) Empirical coefficients and comparison

The empirical coefficients were computed in MATLAB using

$$a_k \approx \text{mean}(x(t_n)e^{-jk\omega_0 t_n})$$

where one period was sampled uniformly with 6000 points.

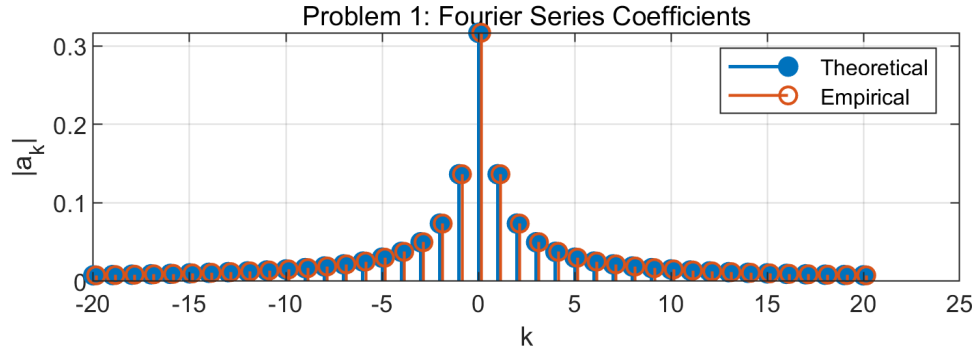


Figure 2: Problem 1 theoretical and empirical coefficient magnitudes.

The numerical coefficients match the theoretical result very closely.

1.4) Reconstruction and the effect of N

The truncated reconstruction is

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$$

As N increases, the approximation improves in the interior of each period. However, because the periodic extension has jump discontinuities at $t = 3m$, oscillations remain near the boundaries.

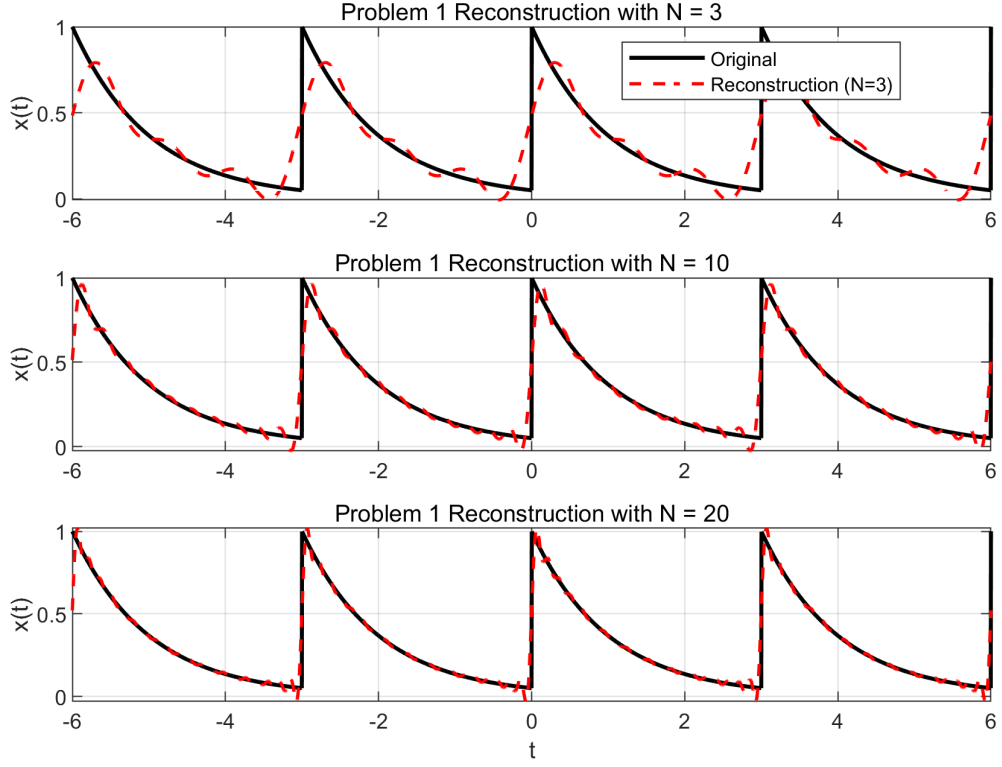


Figure 3: Problem 1 reconstruction for different truncation levels.

In practice, $N = 20$ is sufficient for a good simulation because the shape inside each period is already captured well.

2) Problem 2

The signal is

$$x(t) = \cos\left(\frac{2\pi}{T}t\right) + \left|\cos\left(\frac{2\pi}{T}t\right)\right|, \quad T = 2$$

Hence

$$\omega_0 = \frac{2\pi}{T} = \pi, \quad x(t) = \cos(\pi t) + |\cos(\pi t)|$$

2.1) Plot of the signal

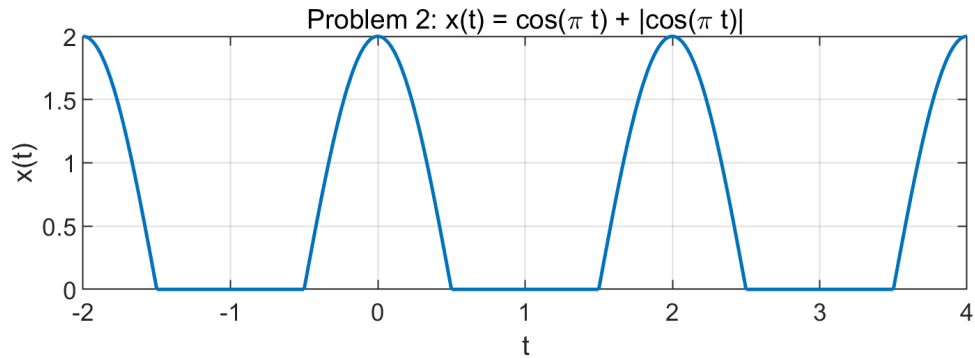


Figure 4: Problem 2 signal plotted over multiple periods.

2.2) Empirical Fourier series coefficients and comparison

The empirical coefficients were obtained by the same averaging method used in Problem 1. They were compared with the theoretical coefficients from HW5:

$$a_0 = \frac{2}{\pi}, \quad a_{\pm 1} = \frac{1}{2}$$

$$a_{\pm 2m} = \frac{2(-1)^{m-1}}{\pi(4m^2 - 1)}, \quad m \geq 1, \quad a_k = 0 \text{ for odd } |k| > 1$$

The match is essentially exact.

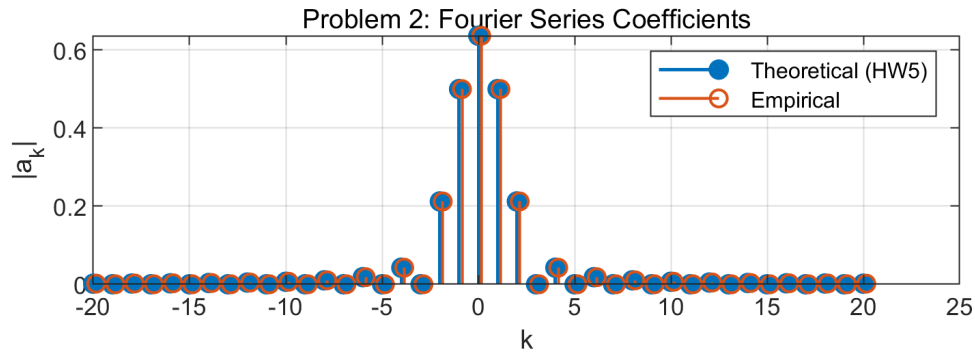


Figure 5: Problem 2 theoretical and empirical coefficient magnitudes.

2.3) Reconstruction and the effect of N

The reconstruction again uses

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$$

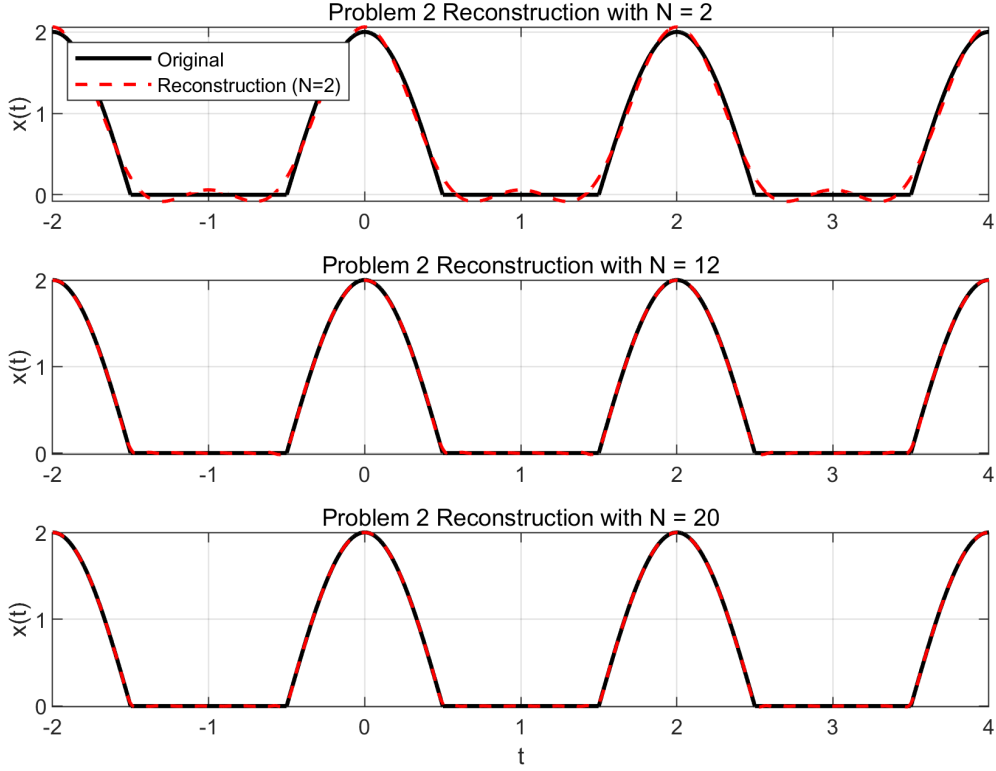


Figure 6: Problem 2 reconstruction for different truncation levels.

Therefore, a sufficient value is approximately $N = 12$.

3) Problem 3

The signal is a periodic triangular waveform with $T = 2$ and one convenient single-period description

$$x(t) = 1 - |t|, \quad -1 \leq t < 1$$

extended periodically with period 2. Since the waveform is even,

$$\omega_0 = \pi$$

and only cosine terms remain in the Fourier series.

3.1) Theoretical Fourier series coefficients

Using the complex exponential formula,

$$a_k = \frac{1}{T} \int_{-1}^1 (1 - |t|) e^{-jk\omega_0 t} dt = \int_0^1 (1 - t) \cos(k\pi t) dt$$

For $k = 0$,

$$a_0 = \int_0^1 (1 - t) dt = \frac{1}{2}$$

For $k \neq 0$,

$$a_k = \int_0^1 (1-t) \cos(k\pi t) dt = \frac{1 - (-1)^k}{\pi^2 k^2}$$

So the coefficient sequence can be written as

$$a_k = \begin{cases} \frac{1}{2}, & k = 0 \\ \frac{2}{\pi^2 k^2}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}$$

3.2) Empirical coefficients and comparison

The empirical coefficients were computed from uniform samples on one period using the MATLAB `mean` function. The result agrees with the theoretical formula to numerical precision.

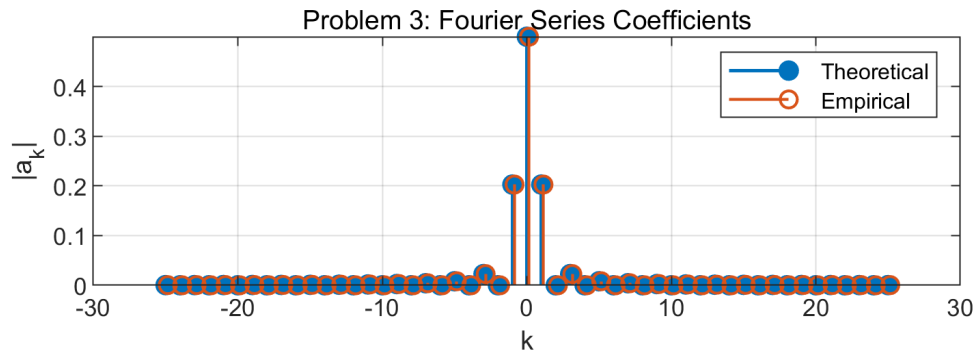


Figure 7: Problem 3 theoretical and empirical coefficient magnitudes.

3.3) Reconstruction and the effect of N

The truncated reconstruction is

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$$

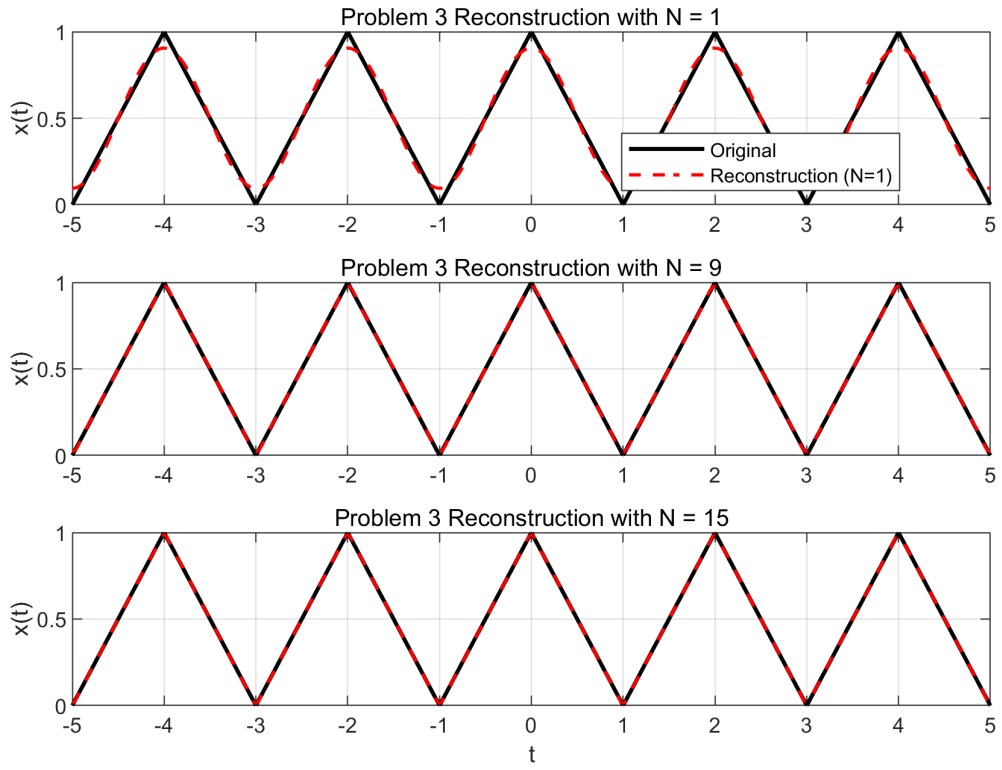


Figure 8: Problem 3 reconstruction for different truncation levels.

So a sufficient choice is $N = 9$.